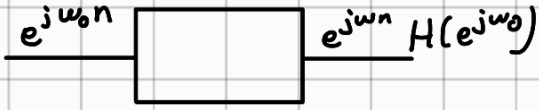
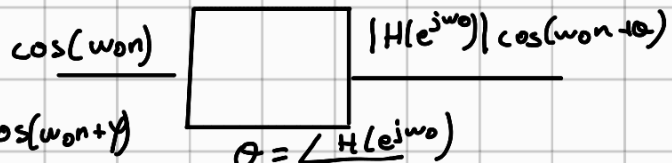


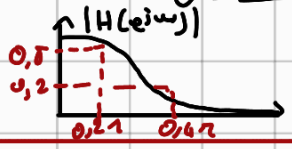
$H(e^{jw}) = |H(e^{jw})|\, e^{j\theta}$

$\theta = \angle H(e^{jw})$





$\cos(w_0 n) = \frac{1}{2} e^{jw_0 n} + \frac{1}{2} e^{-jw_0 n}$



$y[n] = \frac{1}{2} e^{jw_0 n} H(e^{jw_0}) + \frac{1}{2} e^{-jw_0 n} H(e^{-jw_0})$

$h[n]$ reel $\Rightarrow H(e^{jw}) = H^*(e^{-jw})$

$H(e^{-jw}) = H^*(e^{jw})$

$h[n]$ reel $\Rightarrow y[n] = \frac{1}{2} e^{jw_0 n} H(e^{jw_0}) + \frac{1}{2} e^{-jw_0 n} H^*(e^{jw_0})$

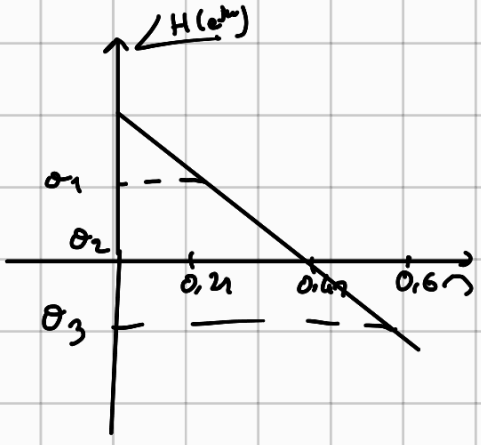
$= \frac{1}{2} e^{jw_0 n} |H(e^{jw_0})| e^{j\theta} + \frac{1}{2} e^{-jw_0 n} |H(e^{jw_0})| e^{-j\theta}$

$= |H(e^{jw_0})| \left[\frac{e^{j(w_0 n + \theta)} + e^{-j(w_0 n + \theta)}}{2} \right]$

$= |H(e^{jw_0})| \cos(w_0 n + \theta)$

$x[n] = \cos(0,2\pi n) + \cos(0,4\pi n) + \cos(0,6\pi n)$

$y[n] = 0,6 \cos(0,2\pi n + \theta_1) + 0,2 \cos(0,4\pi n + \theta_2) + 0 \cdot \cos(0,6\pi n + \theta_3)$



$|H(e^{j\frac{\pi}{2}})| \cos(\frac{\pi}{2} n + \angle H(e^{j\frac{\pi}{2}}))$

$x[n] = \cos(\pi/2\, n)$

$h[n] = \delta[n] - \delta[n-1]$

$y[n]$?

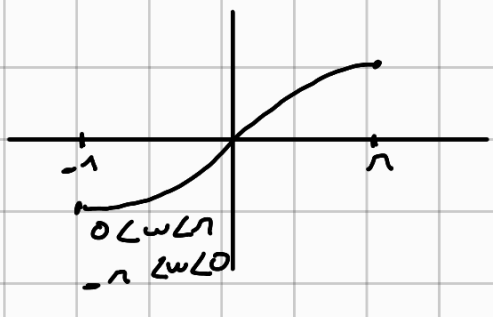
sakin ha $y[n] = \cos(\frac{\pi}{2} n) - \cos(\frac{\pi}{2}(n-1))$

$H(e^{jw}) = 1 - e^{-jw} = 2j \, \frac{e^{-j\frac{w}{2}} (e^{j\frac{w}{2}} - e^{-jw/2})}{2j}$

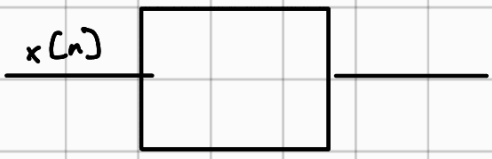
$= j e^{-\frac{w}{2}} 2 \sin(w/2)$

$|H(e^{jw})| = 2 \left| \sin(\frac{w}{2}) \right|$

$\angle H(e^{jw}) = \begin{cases} \frac{\pi}{2} - \frac{w}{2} & 0 < w < \pi \\ \frac{\pi}{2} - \frac{w}{2} + \pi & -\pi < w < 0 \end{cases}$



$$y[n] = \sqrt{2}\cos\left(\frac{\pi}{2}n + \frac{\pi}{4}\right)$$



$$y[n] = \frac{1}{3}x[n] + \frac{1}{3}x[n-1] + \frac{1}{3}x[n-2]$$

$$h[n] = \frac{1}{3}\delta[n] + \frac{1}{3}\delta[n-1] + \frac{1}{3}\delta[n-2]$$

$$H(e^{j\omega}) = \frac{1}{3} + \frac{1}{3}e^{-j\omega} + \frac{1}{3}e^{-j2\omega}$$

$$= \sum_{k=0}^{2} \frac{1}{3}e^{-j\omega k}$$

★ $$\sum_{k=N_1}^{N_2} \alpha^k = \frac{\alpha^{N_1} - \alpha^{N_2+1}}{1-\alpha}$$

$$= \frac{1}{3}\,\frac{(e^{-j\omega})^0 - (e^{-j\omega})^{2+1}}{1 - e^{-j\omega}}$$

$$= \frac{1}{3}\,\frac{1 - e^{-j3\omega}}{1 - e^{-j\omega}} = \frac{1}{3}\,\frac{e^{-j\frac{3}{2}\omega}\left(e^{j\frac{3}{2}\omega} - e^{-j\frac{3}{2}\omega}\right)}{e^{-j\frac{\omega}{2}}\left(e^{j\frac{\omega}{2}} - e^{-j\frac{\omega}{2}}\right)}$$

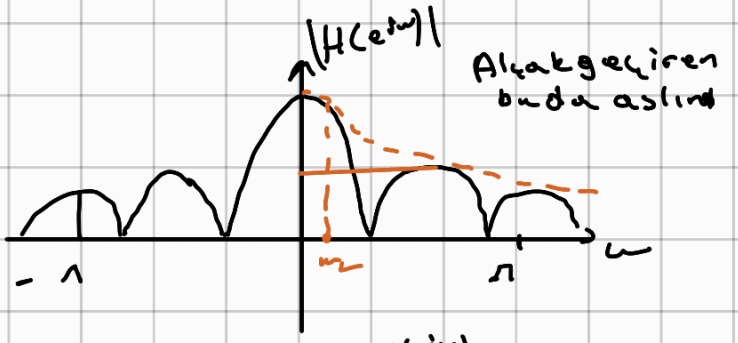
ikisini eşiye bölsek

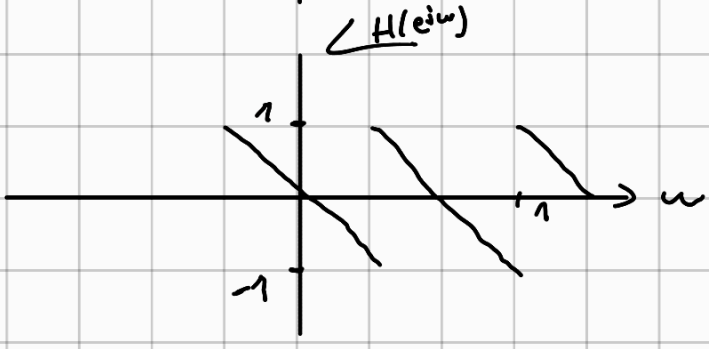
$$= \frac{1}{3}e^{-j\omega}\,\frac{\sin\left(\frac{3}{2}\omega\right)}{\sin(\omega/2)}$$

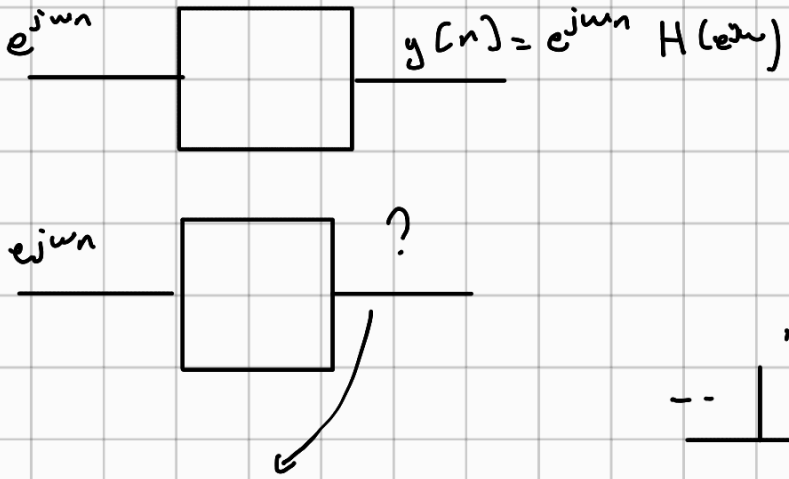
$$|H(e^{j\omega})| = \frac{1}{3}\left|\frac{\sin\frac{3}{2}\omega}{\sin\frac{\omega}{2}}\right|$$

Her zaman sin hale getir

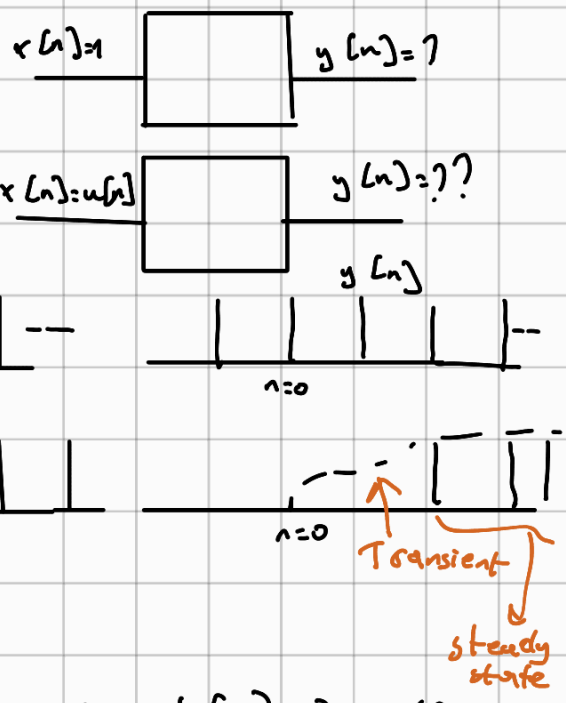
$$\angle H(e^{j\omega}) = -\omega + \pi$$ negatif olduğunda







$$y[n] = \begin{cases} 0 & n < 0 \\ ? & n \geq 0 \end{cases}$$



$$y[n] = \sum_k h[k]\, x[n-k]$$

system is causal $\Rightarrow h[n] = 0 \quad n<0$

$$= \sum_{k=-\infty}^{\infty} h[k]\, e^{j\omega(n-k)}\, u[n-k]$$

$$= \sum_{k=0}^{\infty} h[k]\, e^{j\omega(n-k)}\, u[n-k]$$

$h[n]$ causal

$u[n-k] = 0, \quad n-k<0, \quad k>n$

$$y[n] = \sum_{k=0}^{\infty} h[k]\, e^{j\omega(n-k)} = \sum_{k=0}^{\infty} h[k]\, e^{j\omega k}\, e^{j\omega n} - \sum_{k=n+1}^{\infty} h[k]\, e^{-j\omega k}\, e^{j\omega n}$$

$$= \left(e^{j\omega n} H(e^{j\omega})\right) - \sum_{k=n+1}^{\infty} h[k]\, e^{-j\omega k}\, e^{j\omega n}$$

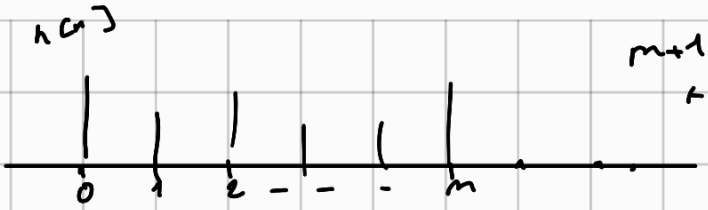
steady state component

$$y[n] = y_{ss}[n] + y_{tr}[n]$$

$$y[n] = y_{ss}[n] + y_{tr}[n]$$

$$y_{ss}[n] = e^{j\omega n} H(e^{j\omega})$$

$$y_{tr}[n] = -\sum_{k=n+1}^{\infty} h[k]\, e^{-j\omega k}\, e^{j\omega n}$$



Filtrenin genişliği m+1

$$y_{tr}[0] = -\sum_{k=1}^{\infty} (\quad) \neq 0$$

$$y_{tr}[1] = -\sum_{k=2}^{\infty} (\quad) \neq 0$$

$$y_{tr}[2] = -\sum_{k=3}^{\infty} (\quad) \neq 0$$

$$\vdots$$

$$y_{tr}[m-1] = -\sum_{k=m+1}^{\infty} (\quad) \neq 0$$

$$y_{tr}[m] = \sum_{k=m+1}^{\infty} (\quad) = 0$$

$$y_{tr}[m+1] = \sum_{k=m+n}^{\infty} (\quad) = 0$$

pencere genişliği -1'e kadar 0dan farklı

pencere genişliği ne kadar dar olursa 0 kadar pencere genişliği dar olur

$$h[n] = \frac{1}{3}\delta[n] + \frac{1}{3}\delta[n-1] + \frac{1}{3}\delta[n-2]$$

$x[n] = 1 \cdot u[n]$

$x[n] = 1$ olsaydı $y[n] = 1$ olurdu

$y[n] = 0 \quad n<0$

$y[0] = \frac{1}{3}$

$y[1] = \frac{2}{3}$

$y[2] = 1$

$y[3] = 1$

} Transient süreç ikiörnek

pencere genişliği eksi 1'i kadar

Peki transient kararlı mı??

$$\left|y_{tr}[n]\right| = \left| \sum_{k=n+1}^{\infty} h[k]\, e^{-j\omega k}\, e^{j\omega n} \right| \leq \sum_{k=n+1}^{\infty} |h[k]|\, |e^{-j\omega k}\, e^{j\omega n}|$$

$$\leq \sum_{k=n+1}^{\infty} |h[k]| \leq \sum_{k=0}^{\infty} |h[k]|$$

Sistem kararlı ise transienti de kararlıdır.

3. bölüm

$$X(e^{jw}) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-jwn}$$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]\, z^{-n}$$

$z = re^{jw}$

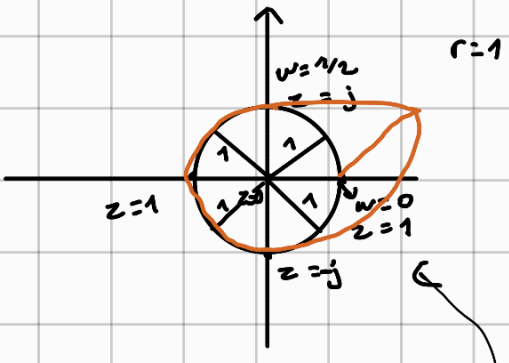
$$= \sum_{n=-\infty}^{\infty} x[n]\,(re^{jw})^{-n}$$

$$= \sum_{n=-\infty}^{\infty} [x[n] r^{-n}]\, e^{-jwn}$$

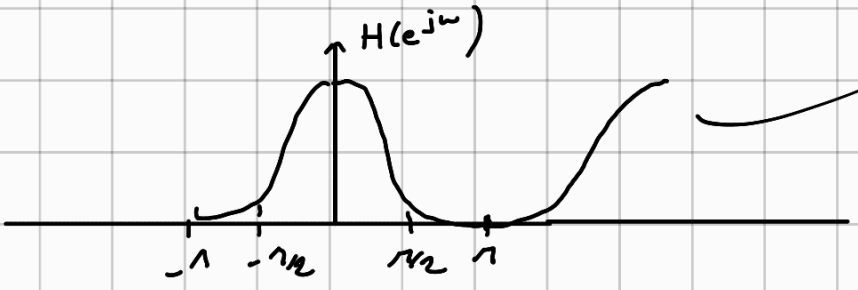
z dönüşümü $x[n]r^{-n}$ 'in Fourier Transform kararlılık analizi için kullanılır

$z = re^{jw}$





r=1 için z transformu ile z dönüşümü aynıdır



Sonuç birim çemberin üzerindeyse sistem kararlıdır

$x[n] \rightarrow X(z)$

$h[n] \rightarrow H(z)$

$y[n] \rightarrow Y(z)$

$$Y(z) = \frac{P(z)}{Q(z)}$$

rasyonel

P(z)'nin kökleri X(z)'nin sıfırları

Q(z)'nin kökleri kutupları

Ex 1 $x[n] = a^n u[n]$, $X(z) = ?$

$$X(z) = \sum_{n=-\infty}^{\infty} a^n u[n] \cdot z^{-n} = \sum_{n=-\infty}^{\infty} \left(\frac{a}{z}\right)^n u[n] = \frac{1}{1-\frac{a}{z}} \qquad \left|\frac{a}{z}\right| < 1 \quad |z| > |a|$$

$\sum_{n=0}^{\infty} a^n$

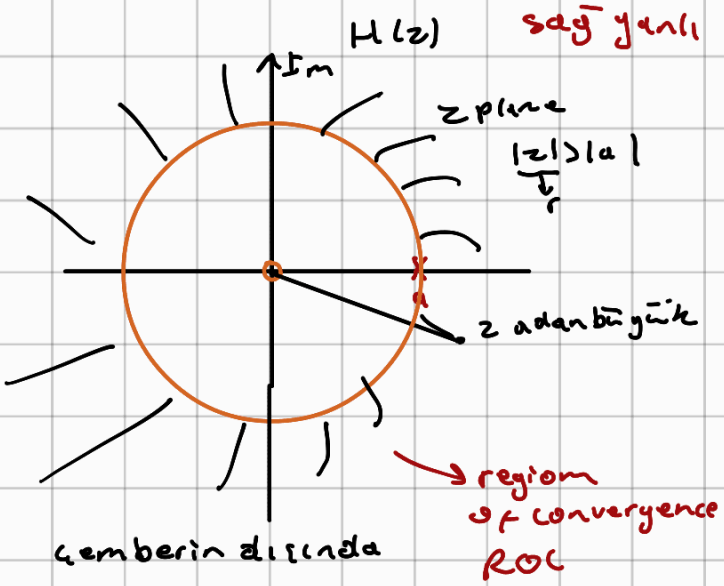
$$X(z) = \frac{1}{1-az^{-1}}$$

$$X(z) = \frac{1}{1-az^{-1}} \rightarrow \text{negatif form}$$

$$= \frac{z}{z-a}$$

mutlaka bu forma getir.

sıfırda bir sıfırı var bağlı göremeol



EX-2 $x[n] = -a^n u[-n-1]$

$$X(z) = \sum_{n=-\infty}^{\infty} -a^n u[-n-1] z^{-n}$$

$$= \sum_{n=-\infty}^{-1} -\left(\frac{a}{z}\right)^n = \sum_{n=1}^{\infty} -\left(\frac{a}{z}\right)^{-n} = \sum_{n=0}^{\infty} -\left(\frac{a}{z}\right)^{-n} - [-1]$$

$$= 1 - \sum_{n=0}^{\infty} \left(\frac{z}{a}\right)^n$$

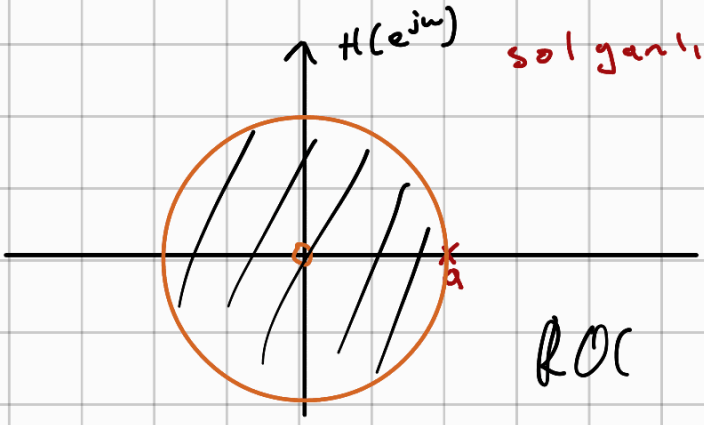
$$= 1 - \frac{1}{1-\frac{z}{a}} \qquad |a| < 1 \quad \left|\frac{z}{a}\right| < 1$$

$$= 1 - \frac{a}{a-z} = \frac{-z}{a-z}$$

$$= \frac{z}{z-a} \qquad \left|\frac{z}{a}\right| < 1 \quad |z| < a$$

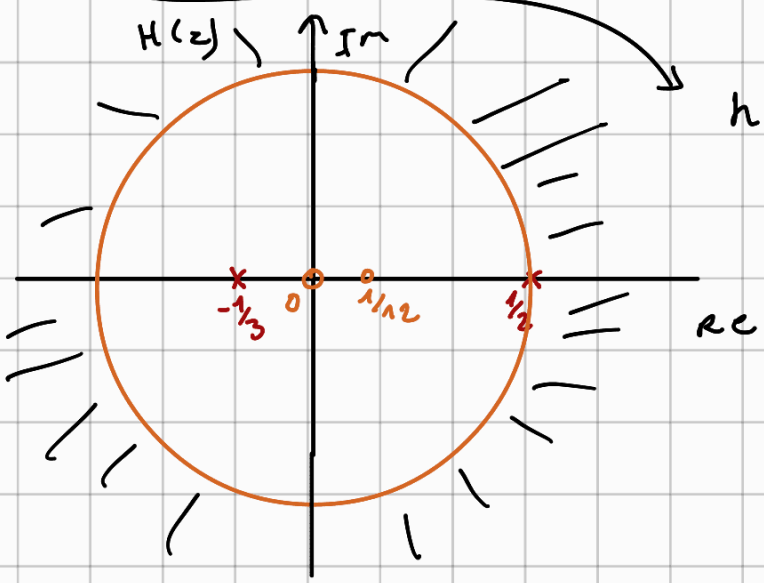
$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a} \quad |a| < 1$$

z dönüşümü ROC'la birlikte anlamlıdır



Ex 3 $x[n] = \left(\frac{1}{2}\right)^n u[n] + \left(-\frac{1}{3}\right)^n u[n]$

$$X(z) = \underbrace{\frac{z}{z-\frac{1}{2}}}_{|z|>1/2} + \underbrace{\frac{z}{z+\frac{1}{3}}}_{|z|>1/3} = \frac{2z\left(z-\frac{1}{12}\right)}{\left(z-\frac{1}{2}\right)\left(z+\frac{1}{3}\right)} \qquad |z|>\frac{1}{2}$$



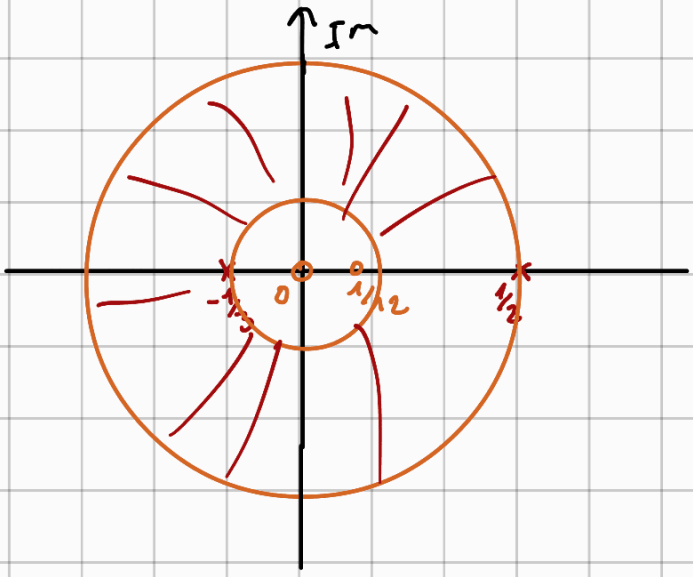
heriki koşul sağlanmalı.

$|z|>1/2$

Ex 4 $x[n] = \left(-\frac{1}{3}\right)^n u[n] - \left(\frac{1}{2}\right)^n u[-n-1]$

$$X(z) = \underbrace{\frac{z}{z+\frac{1}{3}}}_{|z|>1/3} + \underbrace{\frac{z}{z-\frac{1}{2}}}_{|z|<1/2} = \frac{2z\left(z-\frac{1}{12}\right)}{\left(z-\frac{1}{2}\right)\left(z+\frac{1}{3}\right)}$$

$$\frac{1}{3} < |z| < \frac{1}{2}$$



Ex 5 $x[n] = \left(-\frac{1}{2}\right)^n u[n] - \left(\frac{1}{3}\right)^n u[-n-1]$

$$X(z) = \frac{z}{z+1/2} + \frac{z}{z-1/3}$$

$|z| > 1/2$ $\quad$ $|z| < 1/3$

ROC yok

hiç bir zaman yakınsamaz

Ex 6: $x[n] = \begin{cases} a^n & 0 \le n \le N-1 \\ 0 & 0.w. \end{cases}$

$$X(z) = \sum_{0}^{N-1} a^n z^{-n} = 1 + a z^{-1} + a^2 z^{-2} + \cdots + a^{N-1} z^{-(N-1)}$$

$$= 1 + \frac{a}{z} + \frac{a^2}{z^2} + \cdots + \frac{a^{N-1}}{z^{N-1}}$$

$z=0$ 'da toplam sonuz

z hariç her yerde z dönüşümü vardır

$$\frac{1 - a^N}{1 - a z^{-1}}$$

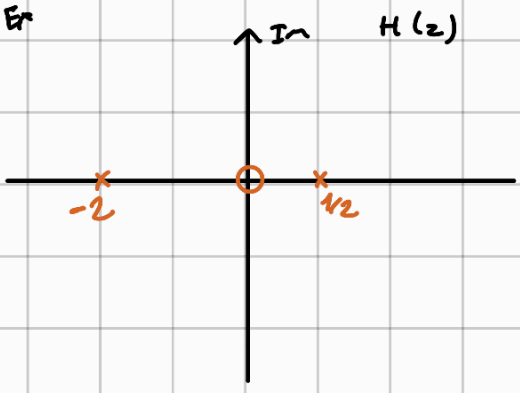
Ex.7

$x[n] = \begin{cases} a^n & -N \le n \le 0 \\ 0 & 0.w \end{cases}$ $\quad X(z) = 1 + a^{-1}z + a^{-2}z^2 + \cdots + a^{-N}z^N$

$z = \infty$ için ∞ olur

$z \neq \infty$ için z dönüşümümüz

Ex



kökcal'in k'sı (+)

$$H(z) = K \frac{z}{(z-\frac{1}{2})(z+2)}$$

$|z| > 2$ ✓ → causal

karlı ise ROC birim çemberi içermeli

$\frac{1}{2} < |z| < 2$ ✓ stable

$|z| < \frac{1}{2}$ ne nedensel ne kararlıdır.

* Sorunun için nedensel derse sağ yanlıdır